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FORM VI

MATHEMATICS

2/3 UNIT

AM WEDNESDAY 17 AUGUST
TRIAL H.S.C. 1988.
TIME; 3 HOURS

200 copies

INSTRUCTIONS:

- * ALL questions may be attempted, and are of equal value.
- * ALL necessary working should be shown in every question. Marks may not be awarded for careless or badly arranged work.
- * Standard integrals are printed on the last page which may be removed.
- * Silent, non-programmable calculators may be used.
- * Each question attempted is to be returned in a separate booklet clearly marked Question 1, Question 2, etc. on the cover. Each booklet must show your Candidate's Number.
- * If you do not attempt a question, you must still hand in a booklet for that question, with NOT ATTEMPTED written clearly on the front.

* * * * *

QUESTION 1: (use a separate book)

(i) Express 493.461

- a) in scientific notation
- b) correct to the nearest tenth

(ii) Simplify $2(2x - 5) - (4 - x)$

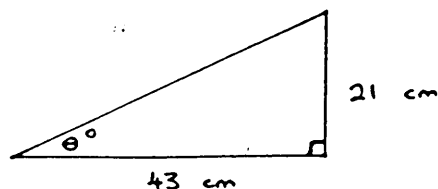
(iii) Solve for x:

a) $\frac{x - 1}{3} = \frac{2x + 5}{2}$

b) $5 - 3x < 8$

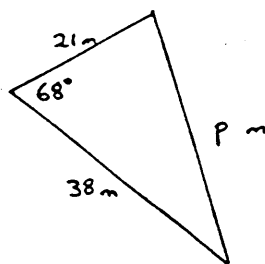
(iv) If $s = ut + \frac{1}{2}at^2$, find s if $u = 15.2$, $t = 3$, and $a = 9.8$

(v) Find the value of θ in the diagram (correct to the nearest degree).



QUESTION 2: (use a separate book)

- (i) Solve $|3x - 5| = 7$
- (ii) Find the product of 4×10^{-3} and 3.2×10^5 , expressing your answer in scientific notation
- (iii) Rationalise the denominator: $\frac{1}{2 - \sqrt{3}}$
- (iv) Factorise fully: $5m^3 + 40$
- (v) Find the value of p in the diagram, correct to 2 decimal places



- (vi) Solve $3^x = 11$, correct to 3 decimal places.

QUESTION 3: (use a separate book)

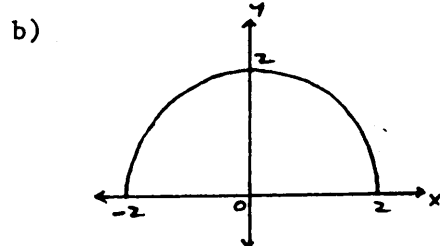
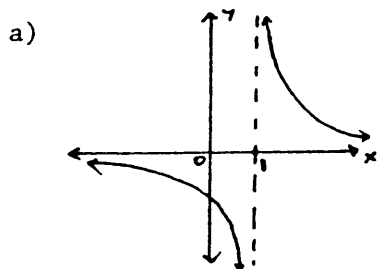
- (i) Find the domain of the function $F(x) = \sqrt{x - 5}$
- (ii) The numbers $x + 3$, $6x$, 18 are in geometric progression. Find the value(s) of x .
- (iii) Differentiate the following expressions with respect to x :
 - a) $x^3 - 3x$
 - b) $\frac{2}{\sqrt{x}}$
 - c) $\cos(3x - 2)$
 - d) $\frac{\log_e x}{x}$

QUESTION 4: (use a separate book)

- (i) Factorise fully: $x^3 - x^2 - x + 1$
- (ii) Find primitives of the following functions (with respect to x)
- a) $\sec^2 3x$
- b) $\frac{1}{\sqrt{3x-5}}$
- (iii) The third and seventh term of an arithmetic progression are 15 and 59 respectively. Find:
- a) the first term and common difference
- b) the sum of the first 18 terms
- (iv) Comment on the validity of the following statement:
"The first term of a geometric progression is 5 and its sum to infinity is 2".

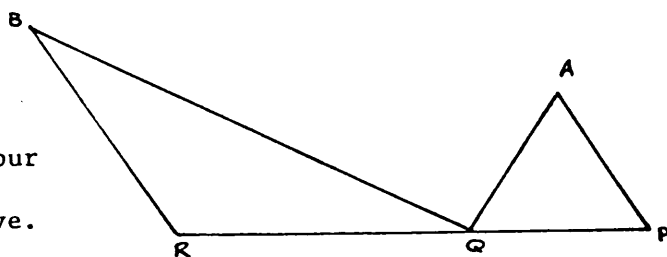
QUESTION 5: (use a separate book)

- (i) Write down possible equations for the following curves:



- (ii) Evaluate $\lim_{x \rightarrow 3} \left(\frac{x^2 - 2x - 3}{x - 3} \right)$

- (iii) In the diagram PQR is a straight line,
PA = PQ, RB = RQ and AP || BR



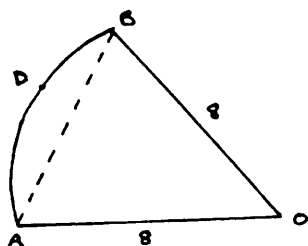
- a) Reproduce the diagram in your answer book, showing on it the features mentioned above.
- b) Prove (giving reasons) that $\hat{AQB} = 90^\circ$
- (iv) ABCD is a quadrilateral whose diagonals bisect each other. Use congruent triangles to prove that $AB \parallel DC$

QUESTION 6: (use a separate book)

- (i) Find the equation of the normal to the curve $y = \sqrt{x}$ at the point where $x = 4$ (give your answer in general form).
- (ii) Consider the expression $p(x) = 2x^2 + 2x + k + 3$
 - a) For what values of k does this expression have real roots?
 - b) For what value(s) of k is the sum of the roots twice the product of the roots?
- (iii) Find the centre and radius of the circle $x^2 + 4x + y^2 - 2y = 11$
- (iv) If $\cos x = \tan x$, find the values of $\sin x$

QUESTION 7: (use a separate book)

- (i) For the parabola $(y - 2)^2 = 2x + 6$ find:
 - a) the vertex
 - b) the focal length
 - c) the equation of the directrix
- (ii) The gradient function of a curve is given by $y' = 3x^2 - 4$. If the curve passes through the point $(-1, 5)$, find the equation of the curve.
- (iii) Find the equation of the locus of all points equidistant from the points $(-2, 1)$ and $(2, 3)$.
- (iv)



The diagram shows a sector of a circle, centre O , whose radius is 8 cm. If the area of the triangle AOB is 25 cm^2 , and $\hat{AOB}$ is acute, find:

- a) the size of $\hat{AOB}$ (in radians, correct to two decimal places)
- b) the length of the arc ADB , correct to the nearest millimetre.

QUESTION 8: (use a separate book)

- (i) Find the EXACT value of $\int_3^4 \frac{x}{x^2 - 8} dx$ in its simplest form
- (ii) a) Find the co-ordinates of the points of intersection of the two curves $y = 4x - x^2 - 3$ and $y = x^2 - 2x + 1$
- b) Calculate the area contained by the two curves between the points of intersection.
- (iii) Frank decides he wants to retire in 25 years time and collect superannuation totalling \$500 000. He decides to invest a set sum of money in a fund which offers an interest rate of 14% per annum (compounded quarterly). If he makes his first payment now, and his last payment 3 months before he retires, what sum of money will he have to invest each quarter to realise his ambition (give your answer to the nearest dollar)?

QUESTION 9: (use a separate book)

- (i) a) Expand and simplify $(x + \frac{1}{x})^2$
- b) The area between the curve $y = x + \frac{1}{x}$, the x-axis and the ordinates $x = 1$ and $x = 3$ is rotated about the x-axis. Find the exact volume of the solid formed.
- c) What is the size of the error if this volume is calculated using the trapezoidal rule with 3 ordinate values (give your answer as a fraction of π)?
- ii) A 4 metre piece of wire is cut into 3 pieces, which are bent to form a square and two congruent circles.
- a) If the radius of each circle is r metres, show that the total area (A square metres) of the 3 figures is given by
- $$A = 2\pi r^2 + (1 - \pi r)^2$$
- b) Find the value of r and the length of the side of the square which will make this total area a minimum (give your answers in EXACT form).

QUESTION 10: (use a separate book)

- (i) a) Sketch the curve $y = 3 \sin 2x$, $0 \leq x \leq 2\pi$
- b) Solve the equation $\sin 2x = \frac{1}{3}$, $0 \leq x \leq 2\pi$, given that the smallest solution in this domain is 0.17 (give all answers correct to two decimal places) {HINT: use your sketch in part (a)}
- (ii) a) Find the stationary point(s) for the curve $y = xe^x$ and determine its (their) nature.
- b) Sketch the curve $y = xe^x$, showing stationary point (s), intercepts and asymptotes (there is no need to find points of inflexion)
- c) Hence or otherwise find the value(s) of k for which the equation $xe^x = k$ has
- α) 2 solutions
- β) 1 solution

E N D O F P A P E R

① i) a) 4.93461×10^2

b) 493.5

ii) $2(2x-5) - (4-x) = 4x-10-4+x = 5x-14$

iii) a) $\frac{x-1}{3} = \frac{2x+5}{2}$ b) $5-3x < 8$
 $6x+15 = 2x-2$ $-3 < 3x$
 $4x = -17$ $-\frac{3}{3} < x$
 $x = -\frac{17}{4}$ $x > -1$

iv) $S = (15 \cdot 2)(3) + \frac{1}{2}(9 \cdot 8)(3)^2$
 $= 89.7$

v) $\tan \theta = \frac{21}{43}$
 $\theta = 26^\circ$

② i) $|3x-5| = 7$

$3x-5 = 7$ or $3x-5 = -7$
 $3x = 12$ or $3x = -2$
 $x = 4$ or $x = -\frac{2}{3}$

ii) $4 \times 10^{-3} \times 3.2 \times 10^5 = 12.8 \times 10^2$
 $= 1.28 \times 10^3$

iii) $\frac{1}{2-\sqrt{2}} \cdot \frac{2+\sqrt{2}}{2+\sqrt{2}} = \frac{2+\sqrt{2}}{4-2}$
 $= 2+\sqrt{2}$

iv) $5m^2 + 40 = 5(m^2 + 2^2)$
 $= 5(m+2)(m^2-2m+4)$

v) $p^2 = 36^2 + 21^2 - 2 \cdot 21 \cdot 36 \cos 68^\circ$
 $p = 35.88$

vi) $2^x = 11$
 $\log_2 11 = x$
 $x = \frac{\log_{10} 11}{\log_{10} 2} = 2.183$

③ i) $x-5 > 0, x \geq 5$

ii) $\frac{6x}{x+3} = \frac{18}{6x} = \frac{3}{x}$

$\therefore 6x^2 = 3x+7$

$2x^2 - x - 3 = 0$

$2x^2 - 3x + 2x - 3 = 0$

$x(2x-3) + 1(2x-3) = 0$

$(x+1)(2x-3) = 0$

$x = -1$ or $\frac{3}{2}$

iii) a) $y = x^3 - 3x$

$y' = 3x^2 - 3$

b) $y = 2x^{-1/2}$

$y' = -x^{-3/2}$

$= -\frac{1}{x\sqrt{x}}$

c) $y = \cos(3x-2)$

$y' = -3 \sin(3x-2)$

d) $y = \frac{\ln x}{x}$

$y' = \frac{x \cdot \frac{1}{x} - \ln x}{x^2}$

$= \frac{1 - \ln x}{x^2}$

④ i) $x^3 - x^2 - x + 1 = x^2(x-1) - 1(x-1)$

$= (x^2-1)(x-1)$

$= (x+1)(x-1)(x-1)$

ii) a) $\int \sec^2 3x \, dx = \frac{1}{3} \tan 3x + c$

b) $\int (3x-5)^{-1/2} \, dx = \frac{(3x-5)^{1/2}}{\frac{1}{2} \cdot 3} + c$
 $= \frac{2\sqrt{3x-5}}{3} + c$

iii) a) $u_3 = 15$ } $a+2d = 15$ } $4d = 44$
 $u_2 = 59$ } $d+6d = 59$ }

$\therefore d = 11, a = -7$

b) $S_n = \frac{n}{2} \{2a-7 + 17 \cdot 11\} = 9 \times 173 = 1557$

iv) $a = 5, \frac{a}{1-r} = 2$

$\therefore \frac{5}{1-r} = 2 \therefore 5 = 2-2r$

$2r = -3$

$r = -1\frac{1}{2}$ ie $|r| > 1$

but S_∞ can't exist if $|r| > 1$

$\therefore$ there is no such sequence.

⑤ i) a) $y = \frac{1}{x-1}$ b) $y = \sqrt{4-x^2}$

ii) $\lim_{x \rightarrow 3} \frac{x^2-2x-3}{x-3} = \lim_{x \rightarrow 3} \frac{(x-3)(x+1)}{x-3}$
 $= \lim_{x \rightarrow 3} (x+1)$
 $= 4$

iii)

b) Label $x = \widehat{BQR}$, $y = \widehat{PQA}$ as shown.

Then $\widehat{BQR} = 180 - 2x$ ($\angle$ sum of $\triangle BQR$)

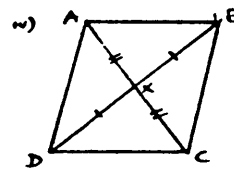
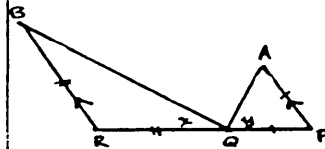
and $\widehat{PQA} = 180 - 2y$ ($\angle$ sum of $\triangle PQA$)

These are const. on $BR \parallel AP$

$\therefore 180 - 2x + 180 - 2y = 180$

$\therefore x + y = 90$

$\therefore \widehat{AOB} = 90^\circ$ (str. $\angle$)



Let the diagonals intersect at X.

In $\triangle ABX$, $\triangle CDX$

$BX = DX$ (AC bisects BD)

$AX = CX$ (BD bisects AC)

$\widehat{AXB} = \widehat{CXD}$ (vert. opp)

$\therefore \triangle ABX \cong \triangle CDX$ (SAS)

$\therefore \widehat{ABX} = \widehat{CDX}$ (corr. $\angle$'s of $\cong \triangle$'s)

$\therefore AB \parallel DC$ (alt. $\angle$'s equal)

⑥ i) $y = x^{1/2}$
 $y' = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}}$

when $x = 4$ $y' = \frac{1}{4}$
 $\therefore m = \frac{1}{4}$

when $x = 4$ $y = 2$

$\therefore$ point is $(4, 2)$

$\therefore$ equation is $y - 2 = \frac{1}{4}(x - 4)$

$y - 2 = \frac{1}{4}x - 1$

$4x + y - 8 = 0$

ii) a) $\Delta \geq 0$ $\Delta = 4 - 4 \times 2 \times (k+3)$

$= 4 - 8(k+3)$

≥ 0

$8(k+3) \leq 4$

$k+3 \leq \frac{1}{2}$

$k \leq -2\frac{1}{2}$

b) $\alpha + \beta = \frac{\pi}{2} = -1$

$\alpha\beta = \frac{k+3}{2}$

$\alpha + \beta = 2\alpha\beta$

$-1 = k+3$

$k = -4$

iii) $x^2 + 4x + y^2 - 2y = 11$

$x^2 + 4x + 4 + y^2 - 2y + 1 = 16$

$(x+2)^2 + (y-1)^2 = 4^2$

$\therefore$ centre = $(-2, 1)$ radius = 4

iv) $\cos x = \frac{\sin x}{\cos x}$

$\cos^2 x = \sin x$

$1 - \sin^2 x = \sin x$

$\sin^2 x + \sin x - 1 = 0$

$\sin x = \frac{-1 \pm \sqrt{1+4}}{2}$

$= \frac{-1 \pm \sqrt{5}}{2}$

⑦) $(y-2)^2 = 2(x+3)$
 $(y-2)^2 = 4 \times \frac{1}{2} (x+3)$
 a) $\sqrt{(-3, 2)}$ b) $\frac{1}{2}$ c) $x = -3\frac{1}{2}$

ii) $y' = 3x^2 - 4$
 $y = x^3 - 4x + c$
 $5 = (-1)^3 - 4(-1) + c$
 $5 = -1 + 4 + c$
 $c = 2 \therefore y = x^3 - 4x + 2$

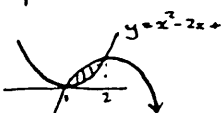
iii) $(x+2)^2 + (y-1)^2 = (x-2)^2 + (y-3)^2$
 $x^2 + 4x + 4 + y^2 - 2y + 1 = x^2 - 4x + 4 + y^2 - 6y + 9$
 $8x + 4y - 8 = 0$
 $2x + y - 2 = 0$

iv) $\frac{1}{2} (8)^2 \sin \theta = 25$
 $\sin \theta = \frac{25}{32}$
 $\theta = 0.90$

b) $\ell = r\theta$
 $= 8 \times 0.90$
 $= 7.2 \text{ cm (nearest mm)}$

⑧) i) $\int_3^4 \frac{x}{x^2-8} dx = \frac{1}{2} \int_3^4 \frac{2x}{x^2-8} dx$
 $= \frac{1}{2} [\ln(x^2-8)]_3^4$
 $= \frac{1}{2} \{ \ln 8 - \ln 1 \}$
 $= \frac{1}{2} \ln 8$
 $= \frac{1}{2} \ln 2^3$
 $= \frac{3}{2} \ln 2$

ii) a) $y = 4x - x^2 - 3$
 $y = x^2 - 2x + 1$
 $4x - x^2 - 3 = x^2 - 2x + 1$
 $0 = 2x^2 - 6x + 4$
 $x^2 - 3x + 2 = 0$
 $(x-2)(x-1) = 0$
 $x = 1 \text{ or } 2$
 $\therefore$ pts of intersection are $(1, 0)$ and $(2, 1)$

b) 
 $A = \int_1^2 (4x - x^2 - 3 - (x^2 - 2x + 1)) dx$
 $= \int_1^2 (6x - 2x^2 - 4) dx$
 $= [3x^2 - \frac{2x^3}{3} - 4x]_1^2$
 $= (12 - \frac{16}{3} - 8) - (3 - \frac{2}{3} - 4)$
 $= 4 - 5\frac{1}{3} + 1 + \frac{2}{3}$
 $= \frac{1}{3}$

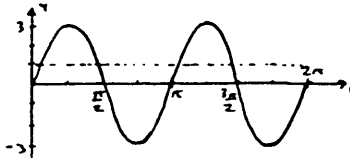
iii) Let m be the amount of each investment:
 1st becomes: $m(1 + \frac{14}{400})^{100} = m(1.035)^{100}$
 2nd " $m(1 + \frac{14}{400})^{99} = m(1.035)^{99}$
 ...
 100th " $m(1 + \frac{14}{400})^1 = m(1.035)^1$
 $\therefore$ Total is $m(1.035) + m(1.035)^2 + \dots + m(1.035)^{100}$
 which is a GP $a = m(1.035)$
 $r = 1.035$
 $n = 100$
 $\therefore$ Total is $\frac{m(1.035)(1.035^{100} - 1)}{0.035} = 500000$
 $\therefore m = \frac{500000 \times 0.035}{1.035(1.035^{100} - 1)} = \560

⑨) a) $(x + \frac{1}{x})^2 = x^2 + 2 \cdot x \cdot \frac{1}{x} + (\frac{1}{x})^2$
 $= x^2 + \frac{1}{x^2} + 2$

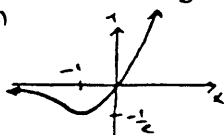
b) $V = \pi \int_1^3 (x + \frac{1}{x})^2 dx$
 $= \pi \int_1^3 (x^2 + \frac{1}{x^2} + 2) dx$
 $= \pi [\frac{x^3}{3} - \frac{1}{x} + 2x]_1^3$
 $= \pi [(9 - \frac{1}{3} + 6) - (\frac{1}{3} - 1 + 2)]$
 $= \frac{40\pi}{3} = 13\frac{1}{3}\pi$

c) $\int_1^3 (x + \frac{1}{x})^2 dx = \frac{1}{2} \{ (1+1)^2 + (3+\frac{1}{3})^2 + 2 \cdot (2+\frac{1}{2})^2 \}$
 $= \frac{1}{2} \{ 4 + \frac{100}{9} + \frac{25}{2} \}$
 $= 13\frac{29}{36}$
 $\therefore V \neq 13\frac{29}{36}\pi$
 $\therefore$ error is $(\frac{29}{36} - \frac{1}{3})\pi = \frac{17}{36}\pi$

ii) 
 $C = 2\pi r$ $C = 2\pi r$ $P = 4\ell$
 $2\pi r + 2\pi r + 4\ell = 4$
 $4(\pi r + \ell) = 4$
 $\therefore \ell = 1 - \pi r$
 $\therefore \text{Area} = \pi r^2 + \pi r^2 + (1 - \pi r)^2$
 $= 2\pi r^2 + (1 - \pi r)^2$
 $A' = 4\pi r + 2(1 - \pi r)(-\pi)$
 $= 4\pi r - 2\pi(1 - \pi r)$
 $= 0 \text{ when } 4\pi r = 2\pi(1 - \pi r)$
 $2r = (1 - \pi r)$
 $r(\pi + 2) = 1$
 $r = \frac{1}{\pi + 2}$
 $A'' = 4\pi + 2\pi^2 > 0$
 $\therefore r = \frac{1}{\pi + 2}$ gives a minimum area
 $\ell = 1 - \pi r = 1 - \frac{\pi}{\pi + 2} = \frac{2}{\pi + 2}$

⑩) i) 
 a) $\sin 2x = \frac{1}{3}$
 $\Leftrightarrow 3 \sin 2x = 1$
 $\therefore$ solve graphically.
 sol^{ns} are:
 $0.17, \frac{\pi}{2} - 0.17, \pi + 0.17$
 a) $\frac{3\pi}{2} - 0.17$
 i.e. $0.17, 1.40, 3.31, 4.54$

ii) a) $y = xe^x$
 $y' = xe^x + e^x$
 $= e^x(x+1)$
 $= 0 \text{ when } x = -1$
 test: $x \quad -2 \quad -1 \quad 0$
 $y' \quad - \quad 0 \quad +$
 $\therefore (-1, \frac{1}{e})$ is a min. t.p.
 when $x=0$ $y=0$
 $\therefore$ curve passes through origin
 as $x \rightarrow -\infty$ $y \rightarrow 0$ $\therefore x$ axis is asymptote on left

b) 
 c) by considering points of intersection with $y=k$, we can see that $xe^x = k$ has
 a) 2 sol^{ns} if $-\frac{1}{e} < k < 0$
 b) 1 solⁿ if $k \geq 0$ or $k = -\frac{1}{e}$